MATRICES

Terminologies

n

m

$$\begin{bmatrix} 4 & 7 & 4 & 2 \\ 6 & 9 & 5 & 5 \\ 3 & 1 & 8 & 0 \\ 1 & 0 & 7 & 2 \end{bmatrix}$$

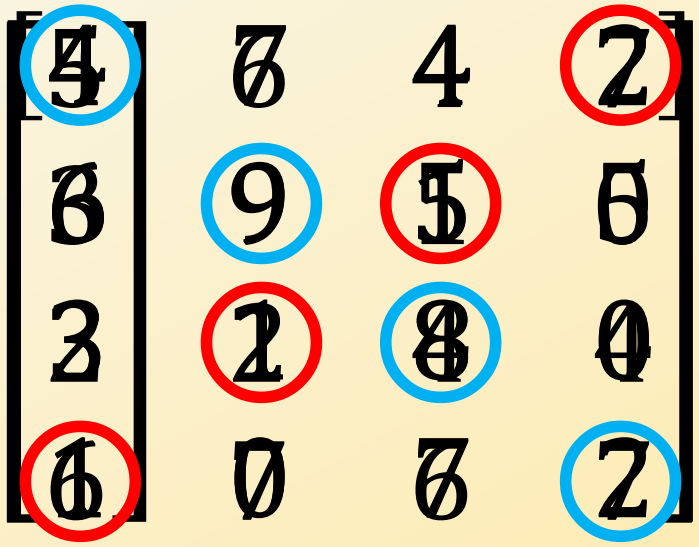
m x n

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}$$

$$a_{11} = 4$$

$$a_{34} = 0$$

Terminologies



Special Matrices

Equal Matrices

$$A = \begin{bmatrix} 2 & 4 & 6 & 8 \\ 0 & 3 & 1 & 3 \\ 1 & 9 & 8 & 0 \\ 0 & 3 & 1 & 4 \end{bmatrix} \qquad B = \begin{bmatrix} 2 & 4 & 6 & 8 \\ 0 & 3 & 1 & 3 \\ 1 & 9 & 8 & 0 \\ 0 & 3 & 1 & 4 \end{bmatrix}$$

$$A \neq B$$

Special Matrices

Square Matrix

$$A = \begin{bmatrix} 2 & 4 & 6 & 8 \\ 0 & 2 & 4 & 3 \\ 8 & 9 & 8 & 0 \\ 0 & 3 & 1 & 4 \end{bmatrix}$$

$m = 4$

$n = 4$

$m = n$

Special Matrices

Zero Matrix

$$\begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 1 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \quad \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Zero-one Matrix

$$\begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 0 \end{bmatrix} \quad \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

Special Matrices

Diagonal

$$\begin{bmatrix} 7 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 3 \end{bmatrix} \quad \begin{bmatrix} 5 & 0 & 0 & 0 \\ 0 & 8 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 3 \end{bmatrix} \quad \begin{bmatrix} 5 & 0 & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 & 0 \\ 0 & 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 4 & 0 \\ 0 & 0 & 0 & 0 & 6 \end{bmatrix}$$

Upper Triangular Diagonal

$$\begin{bmatrix} 7 & 8 & 5 \\ 0 & 4 & 1 \\ 0 & 0 & 3 \end{bmatrix} \quad \begin{bmatrix} 5 & 7 & 9 & 2 \\ 0 & 2 & 3 & 6 \\ 0 & 0 & 1 & 8 \\ 0 & 0 & 0 & 3 \end{bmatrix} \quad \begin{bmatrix} 8 & 0 & 0 & 0 & 0 \\ 5 & 3 & 0 & 0 & 0 \\ 6 & 1 & 2 & 0 & 0 \\ 9 & 6 & 8 & 4 & 0 \\ 2 & 7 & 5 & 7 & 6 \end{bmatrix}$$

Special Matrices

Identity

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$